Assignment 7

NIS3352-01 Modern Cryptography(2)

Marking Rubrics for Assignments.

- You will receive full marks if you idea is (mostly) correct and you haven showed an adequate effort to present your ideas. No marks will be deducted if there are small errors or typos.
- If you are missing an important point of a problem, yet you have partially answered it, 10 marks of the question will be deducted.
- Full marks of the problem will be deducted if (1) your idea is not correct at all or (2) there is an obvious sign that you are copying it from your classmates.
- Each assignment is given one week to finish. Yet, we still accept it if it is at most one week late. For assignments two weeks after the deadline, 10 marks will be deducted.

Problem 1 (25 marks) Let

$$H \in \mathbb{F}_2^{(n-k)\times n}$$

be a parity-check matrix of a linear code. Given a syndrome

$$s \in \mathbb{F}_2^{n-k},$$

the syndrome decoding problem asks for a vector

$$e \in \mathbb{F}_2^n$$

such that

$$He^\top = s \qquad \text{and} \qquad \mathrm{wt}(e) \le t.$$

Now consider the following concrete example:

$$H = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad s = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

1. Find all error vectors e such that

$$He^\top = s, \qquad \mathrm{wt}(e) \le 2.$$

2. Determine whether this syndrome corresponds to a unique minimum-weight error vector, and justify your answer.

Problem 2 (25 marks)

For binary codes, the Gilbert–Varshamov bound has both a finite-length form and an asymptotic form.

Finite-length form. For a binary linear code of length n, dimension k, and minimum distance at least d, the Gilbert–Varshamov criterion implies that such a code exists whenever

$$\sum_{i=0}^{d-2} \binom{n-1}{i} < 2^{n-k}.$$

Asymptotic form. For binary codes, there exist code families with rate R and relative distance δ satisfying

$$R \geq 1 - H_2(\delta),$$

where

$$H_2(x) = -x \log_2 x - (1-x)\log_2(1-x)$$

is the binary entropy function. Consider binary linear codes of length $n = 40$ and dimension $k = 20$.

1. Using the Gilbert–Varshamov criterion

$$\sum_{i=0}^{d-2} \binom{40}{i} < 2^{40-20} = 2^{20},$$

   determine the largest integer d guaranteed by this bound.

2. For rate

$$R = \frac{1}{2},$$

   numerically solve for the asymptotic GV relative distance δ_{GV}, namely the solution of

$$H_2(\delta_{\mathrm{GV}}) = 1 - R = \frac{1}{2},$$

   where

$$H_2(x) = -x \log_2 x - (1-x)\log_2(1-x)$$

   is the binary entropy function.

3. Compare the finite-length conclusion from part (1) with the asymptotic estimate from part (2).

   (*Hint:* for the binomial sums, or more generally for large integer arithmetic, you may use Python, Sage, Mathematica, or another mathematical software package.)

Problem 3 (25 marks) Let $\mathbb{F}$ be a finite field, and let g be a generator of a bilinear group G_1. The public parameters are

$$(g, g^{\tau}, g^{\tau^2}, \ldots, g^{\tau^d}),$$

where $\tau \in \mathbb{F}$ is the secret trapdoor. For a polynomial of degree at most d,

$$f(X) = a_0 + a_1 X + \cdots + a_d X^d,$$

the KZG commitment is defined by

$$C = g^{f(\tau)}.$$

Now consider the polynomial

$$f(X) = 3 + 2X + X^2$$

(with coefficients interpreted in $\mathbb{F}$).

1. Explain how the commitment C to this polynomial is computed from the public parameters.

2. For the point $z = 1$, describe how to construct an opening proof using the quotient polynomial

$$q(X) = \frac{f(X) - f(z)}{X - z},$$

   and explain how the proof π is generated.

3. Write down how the verifier checks the claimed opening using a pairing equation.

4. Prove that if the verification succeeds, then necessarily

$$f(\tau) - f(z) = q(\tau)(\tau - z).$$

   Explain why this identity is the core of KZG correctness.

Problem 4 (25 marks) Let $m_1, \ldots, m_{16}$ be 16 messages, and let H be a collision-resistant hash function with output length λ bits.

Define the leaves by

$$\ell_i = H(m_i), \qquad i = 1, \ldots, 16.$$

Define the first internal level by

$$v_i^{(1)} = H(\ell_{2i-1} \,\|\, \ell_{2i}), \qquad i = 1, \ldots, 8.$$

Define the second level by

$$v_i^{(2)} = H(v_{2i-1}^{(1)} \,\|\, v_{2i}^{(1)}), \qquad i = 1, \ldots, 4.$$

Define the third level by

$$v_i^{(3)} = H(v_{2i-1}^{(2)} \,\|\, v_{2i}^{(2)}), \qquad i = 1, 2.$$

Finally, define the root by

$$r = H(v_1^{(3)} \,\|\, v_2^{(3)}).$$

1. Suppose the prover wants to show that m_{11} belongs to this Merkle tree. Write down the authentication path that must be sent to the verifier.

2. Give the size of this membership proof, and state the following complexities:

   (a) the complexity of building the whole tree,

   (b) the complexity of generating a membership proof for one leaf,

   (c) the complexity of verifying a membership proof for one leaf.